
Portfolio Optimization Engine

*Markowitz Mean-Variance Optimization
and the Black-Litterman Model*

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Abstract

This report presents the theoretical foundations and mathematical derivations underlying the *Portfolio Optimization Engine*—a quantitative tool implementing Markowitz Mean-Variance Optimization (MVO) and the Black-Litterman (BL) model for constructing equity portfolios from NIFTY 50 constituents. We derive the efficient frontier from first principles, establish the Sharpe-ratio maximisation as a tangency problem on the Capital Market Line, and provide a complete Bayesian derivation of the Black-Litterman posterior. Constraints incorporating ESG scores and sector concentration limits are formulated as a constrained non-linear program. Benchmark attribution metrics—Jensen’s alpha, beta, tracking error, and the information ratio—are derived and interpreted. All theoretical claims are referenced to primary sources and standard graduate-level texts.

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1 Introduction and Motivation

Portfolio construction is the central problem of asset management: given n risky assets, choose portfolio weights $\mathbf{w} \in \mathbb{R}^n$ so as to optimise some risk-adjusted return objective. The two dominant frameworks in quantitative equity management are:

- (i) **Mean-Variance Optimization (MVO)** introduced by [Markowitz \[1952\]](#), which solves a quadratic programme to trace the *efficient frontier*;
- (ii) **Black-Litterman (BL) model** [[Black and Litterman, 1990, 1992](#)], developed at Goldman Sachs, which combines equilibrium market returns with subjective analyst views through Bayesian inference.

A key practical failure of raw MVO is that small perturbations in estimated expected returns produce extreme and unstable weight swings. [Michaud and Michaud \[2008\]](#) characterise MVO as an “error maximiser”. Black-Litterman resolves this by anchoring to a stable CAPM equilibrium prior and updating only proportionally to analyst conviction.

The project additionally incorporates ESG-score filtering and sector concentration constraints, and benchmarks portfolio performance against the NIFTY 50 index using standard attribution metrics.

2 Mathematical Notation

Throughout this report the following notation is used.

Symbol	Meaning
n	Number of assets
$\mathbf{w} \in \mathbb{R}^n$	Portfolio weight vector, $\mathbf{1}^\top \mathbf{w} = 1$
$\boldsymbol{\mu} \in \mathbb{R}^n$	Vector of expected annualised returns
$\boldsymbol{\Sigma} \in \mathbb{R}^{n \times n}$	Annualised covariance matrix (positive definite)
R_p, σ_p^2	Portfolio return and variance
R_f	Risk-free rate
δ	Market risk-aversion coefficient
τ	Scalar uncertainty in the prior ($0 < \tau \ll 1$)
$\boldsymbol{\Pi}$	Implied equilibrium excess return vector
$\mathbf{P}$	$k \times n$ view pick matrix
$\mathbf{Q}$	$k \times 1$ view return vector
$\boldsymbol{\Omega}$	$k \times k$ diagonal view-uncertainty matrix
$\boldsymbol{\mu}_{\text{BL}}$	Black-Litterman posterior expected returns

3 Data and Return Computation

3.1 Price Data

Adjusted daily closing prices $S_{i,t}$ for $n = 15$ NIFTY 50 constituents are obtained via Yahoo Finance over the period 1 January 2021 to 31 December 2024 (approximately 986 trading days). Adjusted prices account for dividends and corporate actions [Campbell et al., 1997].

3.2 Simple and Log Returns

The *simple daily return* of asset i on day t is

$$r_{i,t} = \frac{S_{i,t} - S_{i,t-1}}{S_{i,t-1}}. \quad (1)$$

For portfolio construction over moderate holding periods, simple returns are preferred because portfolio returns aggregate linearly:

$$R_p = \mathbf{w}^\top \mathbf{r}_t = \sum_{i=1}^n w_i r_{i,t}. \quad (2)$$

3.3 Annualisation

Sample mean and covariance estimated from daily returns are annualised by

$$\hat{\mu}_i = \bar{r}_i \times 252, \quad (3)$$

$$\hat{\Sigma}_{ij} = \widehat{\text{Cov}}(r_i, r_j) \times 252. \quad (4)$$

The factor 252 reflects the conventional number of trading days per year [Grinold and Kahn, 2000, Ch. 3].

4 Markowitz Mean-Variance Optimization

Primary reference: Markowitz [1952]; see also Luenberger [1998, Ch. 4–6], Fabozzi et al. [2007, Ch. 2].

4.1 Portfolio Mean and Variance

For a long-only portfolio with weight vector $\mathbf{w} \geq \mathbf{0}$, $\mathbf{1}^\top \mathbf{w} = 1$:

$$\mu_p = \mathbb{E}[R_p] = \mathbf{w}^\top \boldsymbol{\mu}, \quad (5)$$

$$\sigma_p^2 = \text{Var}(R_p) = \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}. \quad (6)$$

Definition 4.1 (Efficient Portfolio). *A portfolio $\mathbf{w}^*$ is mean-variance efficient if there exists no portfolio $\mathbf{w}'$ such that $\mu_{p'} \geq \mu_p$ and $\sigma_{p'}^2 \leq \sigma_p^2$ with at least one strict inequality.*

4.2 The Minimum-Variance Frontier

For a target return $\mu^* \in [\mu_{\min}, \mu_{\max}]$, the minimum-variance portfolio solves:

$$\begin{aligned} \min_{\mathbf{w}} \quad & \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} \\ \text{subject to} \quad & \mathbf{w}^\top \boldsymbol{\mu} = \mu^*, \\ & \mathbf{1}^\top \mathbf{w} = 1, \\ & \mathbf{w} \geq \mathbf{0}. \end{aligned} \quad (\text{MVO})$$

4.2.1 Analytical Solution (No Short-Selling Constraint)

Without the no-short-selling constraint, introduce Lagrange multipliers $\lambda, \gamma \in \mathbb{R}$ and form the Lagrangian:

$$\mathcal{L}(\mathbf{w}, \lambda, \gamma) = \frac{1}{2} \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w} - \lambda (\mathbf{w}^\top \boldsymbol{\mu} - \mu^*) - \gamma (\mathbf{1}^\top \mathbf{w} - 1). \quad (7)$$

The first-order conditions are:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{w}} = \Sigma \mathbf{w} - \lambda \boldsymbol{\mu} - \gamma \mathbf{1} = \mathbf{0} \implies \mathbf{w} = \Sigma^{-1}(\lambda \boldsymbol{\mu} + \gamma \mathbf{1}). \quad (8)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = \mathbf{w}^\top \boldsymbol{\mu} - \mu^* = 0. \quad (9)$$

$$\frac{\partial \mathcal{L}}{\partial \gamma} = \mathbf{1}^\top \mathbf{w} - 1 = 0. \quad (10)$$

Define the scalars $A = \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu}$, $B = \boldsymbol{\mu}^\top \Sigma^{-1} \mathbf{1}$, $C = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$, and $D = AC - B^2 > 0$. Substituting (8) into (9)–(10) and solving the 2×2 linear system:

$$\begin{pmatrix} A & B \\ B & C \end{pmatrix} \begin{pmatrix} \lambda \\ \gamma \end{pmatrix} = \begin{pmatrix} \mu^* \\ 1 \end{pmatrix} \implies \lambda = \frac{C\mu^* - B}{D}, \quad \gamma = \frac{A - B\mu^*}{D}. \quad (11)$$

Substituting back gives the optimal weight vector:

$$\boxed{\mathbf{w}^*(\mu^*) = \frac{C\mu^* - B}{D} \Sigma^{-1} \boldsymbol{\mu} + \frac{A - B\mu^*}{D} \Sigma^{-1} \mathbf{1}.} \quad (12)$$

The corresponding minimum portfolio variance is:

$$\sigma_p^{*2}(\mu^*) = \mathbf{w}^{*\top} \Sigma \mathbf{w}^* = \frac{C(\mu^*)^2 - 2B\mu^* + A}{D}. \quad (13)$$

Equation (13) is a parabola in (σ_p^2, μ_p) space, or equivalently a hyperbola in (σ_p, μ_p) space—the *efficient frontier* [Luenberger, 1998, Thm. 4.1].

4.3 Global Minimum Variance Portfolio

The vertex of the parabola (13) is the *global minimum variance portfolio* (GMVP):

$$\mu_{\text{GMVP}} = \frac{B}{C} = \frac{\boldsymbol{\mu}^\top \Sigma^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}, \quad (14)$$

$$\sigma_{\text{GMVP}}^2 = \frac{1}{C} = \frac{1}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}, \quad (15)$$

$$\mathbf{w}_{\text{GMVP}} = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}. \quad (16)$$

With the non-negativity constraint $\mathbf{w} \geq \mathbf{0}$, no closed form exists; we solve (MVO) using Sequential Least Squares Programming (SLSQP) [Kraft, 1988].

4.4 Sharpe Ratio and the Tangency Portfolio

Definition 4.2 (Sharpe Ratio). *For a portfolio with return μ_p and volatility σ_p , the Sharpe ratio is*

$$S(\mathbf{w}) = \frac{\mu_p - R_f}{\sigma_p} = \frac{\mathbf{w}^\top \boldsymbol{\mu} - R_f}{\sqrt{\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}}}. \quad (17)$$

Maximising $S(\mathbf{w})$ subject to $\mathbf{1}^\top \mathbf{w} = 1$, $\mathbf{w} \geq \mathbf{0}$ yields the *tangency portfolio*—the point at which the Capital Market Line (CML) is tangent to the efficient frontier.

4.4.1 Derivation of the Tangency Portfolio (Unconstrained)

Define the excess return vector $\tilde{\boldsymbol{\mu}} = \boldsymbol{\mu} - R_f \mathbf{1}$. Without non-negativity constraints, set $\nabla_{\mathbf{w}} S = \mathbf{0}$:

$$\begin{aligned} \frac{\partial S}{\partial \mathbf{w}} &= \frac{\tilde{\boldsymbol{\mu}}}{\sigma_p} - \frac{(\mathbf{w}^\top \tilde{\boldsymbol{\mu}}) \boldsymbol{\Sigma} \mathbf{w}}{\sigma_p^3} = \mathbf{0} \\ \implies \tilde{\boldsymbol{\mu}} &= \frac{\mathbf{w}^\top \tilde{\boldsymbol{\mu}}}{\sigma_p^2} \boldsymbol{\Sigma} \mathbf{w}. \end{aligned} \quad (18)$$

Since the scalar factor is irrelevant (normalised by the budget constraint), the unnormalised solution is $\tilde{\mathbf{w}} = \boldsymbol{\Sigma}^{-1} \tilde{\boldsymbol{\mu}}$, so:

$$\boxed{\mathbf{w}_{\text{tan}} = \frac{\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - R_f \mathbf{1})}{\mathbf{1}^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - R_f \mathbf{1})}}. \quad (19)$$

The CML passes through $(0, R_f)$ and $(\sigma_{\text{tan}}, \mu_{\text{tan}})$:

$$\mu_{\text{CML}}(\sigma) = R_f + \frac{\mu_{\text{tan}} - R_f}{\sigma_{\text{tan}}} \sigma. \quad (20)$$

Theorem 4.1 (Mutual Fund Separation). *Every efficient portfolio is a convex combination of the risk-free asset and the tangency portfolio. Investors with different risk aversion need hold only two funds.*

Proof sketch: Any portfolio on the CML achieves $\mathbf{w} = \alpha \mathbf{w}_{\text{tan}} + (1 - \alpha) \mathbf{e}_f$ for some $\alpha \in [0, 1]$, where $\mathbf{e}_f$ represents the risk-free asset. The Sharpe ratio of any convex combination cannot exceed $S(\mathbf{w}_{\text{tan}})$ by the quasi-concavity of S [Luenberger, 1998, Ch. 5]. $\square$

5 The Black-Litterman Model

Primary references: Black and Litterman [1990] (original GS internal memo), Black and Litterman [1992]; exposition follows He and Litterman [1999] and Idzorek [2005]. See also Fabozzi et al. [2007, Ch. 9] and Grinold and Kahn [2000, Pt. II].

5.1 Motivation: The Error-Maximisation Problem

Substituting sample estimates $(\hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Sigma}})$ directly into (19) is statistically fragile: $\hat{\boldsymbol{\mu}}$ has standard error of order $\sigma/\sqrt{T}$, which dominates the signal at typical horizons. The resulting weights are extreme, concentrated, and highly sensitive to small changes in inputs [Michaud and Michaud, 2008, Ch. 3].

Black-Litterman solve this by treating expected returns as *random variables* with a prior distribution anchored to CAPM equilibrium, then updating via Bayes' theorem.

5.2 Step 1 — Reverse Optimisation: Implied Equilibrium Returns

Assume the market is in CAPM equilibrium and that the market-capitalisation weight vector $\mathbf{w}_{\text{mkt}}$ is mean-variance optimal. Then $\mathbf{w}_{\text{mkt}}$ satisfies the first-order condition:

$$\boldsymbol{\mu}_{\text{eq}} = \delta \boldsymbol{\Sigma} \mathbf{w}_{\text{mkt}}, \quad (21)$$

where δ is the market risk-aversion coefficient. This is *reverse optimisation*: rather than $(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \rightarrow \mathbf{w}$, we compute $(\boldsymbol{\Sigma}, \mathbf{w}_{\text{mkt}}) \rightarrow \boldsymbol{\mu}_{\text{eq}}$.

We denote the implied excess equilibrium return vector as:

$$\boldsymbol{\Pi} = \delta \boldsymbol{\Sigma} \mathbf{w}_{\text{mkt}}. \quad (22)$$

Remark 5.1. δ is typically calibrated as $\delta = (\mu_{\text{mkt}} - R_f)/\sigma_{\text{mkt}}^2 \approx 2.5$, representing the risk aversion of a representative investor [He and Litterman, 1999].

5.3 Step 2 — Prior Distribution

The Black-Litterman prior on expected returns is:

$$\boldsymbol{\mu} \sim \mathcal{N}(\boldsymbol{\Pi}, \tau \boldsymbol{\Sigma}), \quad (23)$$

where $\tau \in (0, 1)$ is a scalar expressing uncertainty in the prior. The distribution (23) says that the true expected returns are centred at the equilibrium $\boldsymbol{\Pi}$ with covariance $\tau \boldsymbol{\Sigma}$ proportional to the asset covariance. Small τ means strong confidence in equilibrium; typical values are $\tau \in [0.01, 0.10]$ [Idzorek, 2005].

5.4 Step 3 — Investor Views

An analyst specifies k views in the form

$$\mathbf{P} \boldsymbol{\mu} = \mathbf{Q} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Omega}), \quad (24)$$

where:

- $\mathbf{P} \in \mathbb{R}^{k \times n}$ is the *pick matrix*: row j encodes the assets referenced by view j . An absolute view on asset i has $P_{ji} = 1$, all others zero. A relative view (asset i outperforms asset ℓ) has $P_{ji} = +1$, $P_{j\ell} = -1$.
- $\mathbf{Q} \in \mathbb{R}^k$ contains the view expected returns.
- $\mathbf{\Omega} \in \mathbb{R}^{k \times k}$ is a diagonal matrix of view uncertainties: $\Omega_{jj} = \sigma_j^2$ is the variance attached to view j .

This project uses two views:

View 1 (absolute): $P_1 = \mathbf{e}_{\text{TCS}}$, $Q_1 = 0.12$ (12% annual return),

View 2 (relative): $P_2 = \mathbf{e}_{\text{INFY}} - \mathbf{e}_{\text{SBIN}}$, $Q_2 = 0.05$.

5.5 Step 4 — Bayesian Posterior Derivation

Theorem 5.1 (Black-Litterman Posterior). *Given the prior (23) and the view likelihood (24), the posterior distribution of $\boldsymbol{\mu}$ is*

$$\boldsymbol{\mu} \mid \mathbf{Q} \sim \mathcal{N}(\boldsymbol{\mu}_{\text{BL}}, \mathbf{M}^{-1}), \quad (25)$$

where

$$\mathbf{M} = (\tau \boldsymbol{\Sigma})^{-1} + \mathbf{P}^\top \boldsymbol{\Omega}^{-1} \mathbf{P}, \quad (26)$$

$$\boldsymbol{\mu}_{\text{BL}} = \mathbf{M}^{-1} [(\tau \boldsymbol{\Sigma})^{-1} \boldsymbol{\Pi} + \mathbf{P}^\top \boldsymbol{\Omega}^{-1} \mathbf{Q}]. \quad (27)$$

Proof. The prior on $\boldsymbol{\mu}$ is $p(\boldsymbol{\mu}) \propto \exp(-\frac{1}{2}(\boldsymbol{\mu} - \boldsymbol{\Pi})^\top (\tau \boldsymbol{\Sigma})^{-1} (\boldsymbol{\mu} - \boldsymbol{\Pi}))$. The view likelihood is $p(\mathbf{Q} \mid \boldsymbol{\mu}) \propto \exp(-\frac{1}{2}(\mathbf{Q} - \mathbf{P}\boldsymbol{\mu})^\top \boldsymbol{\Omega}^{-1} (\mathbf{Q} - \mathbf{P}\boldsymbol{\mu}))$.

By Bayes' theorem, $p(\boldsymbol{\mu} \mid \mathbf{Q}) \propto p(\mathbf{Q} \mid \boldsymbol{\mu}) p(\boldsymbol{\mu})$. Expanding the exponent and completing the square in $\boldsymbol{\mu}$:

$$\begin{aligned} -2 \ln p(\boldsymbol{\mu} \mid \mathbf{Q}) &= (\boldsymbol{\mu} - \boldsymbol{\Pi})^\top (\tau \boldsymbol{\Sigma})^{-1} (\boldsymbol{\mu} - \boldsymbol{\Pi}) + (\mathbf{Q} - \mathbf{P}\boldsymbol{\mu})^\top \boldsymbol{\Omega}^{-1} (\mathbf{Q} - \mathbf{P}\boldsymbol{\mu}) + \text{const} \\ &= \boldsymbol{\mu}^\top [(\tau \boldsymbol{\Sigma})^{-1} + \mathbf{P}^\top \boldsymbol{\Omega}^{-1} \mathbf{P}] \boldsymbol{\mu} - 2\boldsymbol{\mu}^\top [(\tau \boldsymbol{\Sigma})^{-1} \boldsymbol{\Pi} + \mathbf{P}^\top \boldsymbol{\Omega}^{-1} \mathbf{Q}] + \text{const}. \end{aligned}$$

This is a quadratic in $\boldsymbol{\mu}$, so the posterior is Gaussian with precision matrix $\mathbf{M}$ (equation (26)) and mean $\boldsymbol{\mu}_{\text{BL}} = \mathbf{M}^{-1} [(\tau \boldsymbol{\Sigma})^{-1} \boldsymbol{\Pi} + \mathbf{P}^\top \boldsymbol{\Omega}^{-1} \mathbf{Q}]$ (equation (27)). $\square$

An equivalent, computationally convenient form obtained via the Woodbury matrix identity [Bishop, 2006, A.3] is:

$$\boldsymbol{\mu}_{\text{BL}} = \boldsymbol{\Pi} + \tau \boldsymbol{\Sigma} \mathbf{P}^\top (\tau \mathbf{P} \boldsymbol{\Sigma} \mathbf{P}^\top + \boldsymbol{\Omega})^{-1} (\mathbf{Q} - \mathbf{P} \boldsymbol{\Pi}). \quad (28)$$

Equation (28) makes the Bayesian update transparent: $\boldsymbol{\mu}_{\text{BL}}$ equals equilibrium $\boldsymbol{\Pi}$

plus a correction term proportional to the *view surprise* $(\mathbf{Q} - \mathbf{P}\mathbf{\Pi})$, scaled by analyst confidence $\mathbf{\Omega}^{-1}$.

5.6 Step 5 — View Uncertainty Calibration (Idzorek Method)

Idzorek [2005] proposes setting $\mathbf{\Omega}$ as a diagonal matrix:

$$\Omega_{jj} = \tau (\mathbf{P}\mathbf{\Sigma}\mathbf{P}^\top)_{jj} = \tau \mathbf{p}_j^\top \mathbf{\Sigma} \mathbf{p}_j, \quad (29)$$

where $\mathbf{p}_j^\top$ is the j -th row of $\mathbf{P}$. This scales view uncertainty proportionally to the historical variance of the assets in each view, ensuring well-conditioned matrices and consistent units.

5.7 Step 6 — Posterior Optimization

With posterior mean $\boldsymbol{\mu}_{\text{BL}}$ substituted for $\boldsymbol{\mu}$, the standard Sharpe-ratio maximisation (19) is run:

$$\mathbf{w}_{\text{BL}} = \arg \max_{\mathbf{w}} \frac{\mathbf{w}^\top \boldsymbol{\mu}_{\text{BL}} - R_f}{\sqrt{\mathbf{w}^\top \mathbf{\Sigma} \mathbf{w}}}, \quad \mathbf{1}^\top \mathbf{w} = 1, \mathbf{w} \geq \mathbf{0}. \quad (30)$$

6 Constrained Optimization: ESG and Sector Limits

Reference: Fabozzi et al. [2007, Ch. 7], Roncalli [2013].

6.1 ESG Score Filter

Let $e_i \in [0, 100]$ denote the ESG score of asset i . Assets with $e_i < \underline{e}$ are excluded before optimization. The eligible set is $\mathcal{E} = \{i : e_i \geq \underline{e}\}$ with $|\mathcal{E}| = n' \leq n$.

6.2 Sector Concentration Constraint

Partition the n' eligible assets into G sectors $\mathcal{G}_g \subset \mathcal{E}$, $g = 1, \dots, G$, $\bigcup_g \mathcal{G}_g = \mathcal{E}$. The constrained problem is:

$$\begin{aligned} & \max_{\mathbf{w} \in \mathbb{R}^{n'}} \frac{\mathbf{w}^\top \hat{\boldsymbol{\mu}} - R_f}{\sqrt{\mathbf{w}^\top \hat{\mathbf{\Sigma}} \mathbf{w}}} \\ & \text{subject to } \mathbf{1}^\top \mathbf{w} = 1, \\ & \quad w_i \geq 0 \quad \forall i \in \mathcal{E}, \\ & \quad \sum_{i \in \mathcal{G}_g} w_i \leq \kappa \quad \forall g = 1, \dots, G, \end{aligned} \quad (\text{C-MVO})$$

where $\kappa \in (0, 1)$ is the sector weight cap (set to 0.40 in the implementation). Problem (C-MVO) is a nonlinear programme with linear constraints; it is solved via SLSQP [Kraft, 1988].

Remark 6.1. *The objective function of (C-MVO) is quasi-concave but not concave (the ratio of a linear function to the square root of a convex function); the global optimum is guaranteed for convex feasible sets, which holds here since all constraints are linear [Boyd and Vandenberghe, 2004, Ch. 4].*

7 Performance Attribution and Benchmark Comparison

Reference: Grinold and Kahn [2000, Ch. 6], Fabozzi et al. [2007, Ch. 3].

7.1 CAPM Beta and Jensen's Alpha

The portfolio return is modelled against the benchmark (NIFTY 50) return R_b via the market model:

$$R_p - R_f = \alpha + \beta(R_b - R_f) + \varepsilon, \quad \mathbb{E}[\varepsilon] = 0. \quad (31)$$

OLS estimates give:

$$\hat{\beta} = \frac{\widehat{\text{Cov}}(R_p, R_b)}{\widehat{\text{Var}}(R_b)}, \quad (32)$$

$$\hat{\alpha} = \overline{R_p} - \hat{\beta} \overline{R_b} - (1 - \hat{\beta})R_f. \quad (33)$$

$\hat{\alpha}$ (Jensen's alpha) measures the annualised return in excess of what CAPM predicts given the portfolio's market exposure.

7.2 Tracking Error and Information Ratio

The *active return* on day t is $a_t = R_{p,t} - R_{b,t}$.

$$\text{TE} = \hat{\sigma}(a_t) \times \sqrt{252} \quad (\text{annualised tracking error}), \quad (34)$$

$$\text{IR} = \frac{\bar{a}}{\hat{\sigma}(a_t)} \times \sqrt{252} = \frac{\overline{R_p} - \overline{R_b}}{\text{TE}}. \quad (35)$$

Remark 7.1. *Grinold and Kahn [2000, p. 114] establish the Fundamental Law of Active Management: $\text{IR} \approx \text{IC} \times \sqrt{N}$, where IC is the information coefficient (skill) and N is the number of independent bets. An $\text{IR} > 0.5$ is considered good; $\text{IR} > 1.0$ is exceptional.*

8 Numerical Implementation

8.1 Language Choice: Python 3.10

Python is the dominant language in quantitative finance. Goldman Sachs internal Strats workflows, the SecDB system, and the Marquee platform all use Python as the primary scripting interface [Narang, 2013]. The scientific stack (NumPy, SciPy, Pandas) provides native matrix algebra mapped directly to the linear algebra of Sections 4–5.

8.2 Optimization Solver

SciPy’s `minimize` with `method='SLSQP'` [Kraft, 1988] solves both (MVO) and (C-MVO). SLSQP handles nonlinear objectives with equality and inequality constraints via a successive quadratic programming approach, converging superlinearly near the solution [Nocedal and Wright, 2006, Ch. 18].

8.3 Key Implementation Decisions

Module	Formula implemented	Reference
Return estimation	Eq. (3)–(4)	Grinold and Kahn [2000, Ch. 3]
MVO efficient frontier	Eq. (12)	Markowitz [1952]
Max Sharpe (tangency)	Eq. (19)	Luenberger [1998, Ch. 6]
Reverse optimisation	Eq. (22)	He and Litterman [1999]
BL posterior	Eq. (27)	Black and Litterman [1992]
View uncertainty	Eq. (29)	Idzorek [2005]
Constrained MVO	Eq. (C-MVO)	Roncalli [2013]
Jensen’s alpha	Eq. (33)	Grinold and Kahn [2000, Ch. 6]
Information ratio	Eq. (35)	Grinold and Kahn [2000, p. 114]

9 Results Summary

Portfolio	Ann. Return	Ann. Volatility	Sharpe Ratio	Notes
Max Sharpe (MVO)	29.95%	16.28%	1.840	Tangency portfolio
Min Variance (MVO)	14.68%	12.47%	1.177	GMVP
Black-Litterman	8.60%	20.59%	0.102	TCS/INFY views applied
ESG Constrained	30.91%	18.31%	1.333	$\kappa = 0.40$, $\underline{e} = 65$
NIFTY 50 Benchmark	14.43%	—	—	Passive index
<i>Benchmark attribution (Max Sharpe portfolio):</i>				
Jensen's α : +16.53%		β : 0.890		IR: 1.557

The high information ratio of 1.557 is a backtest artefact—it reflects in-sample optimisation over the same period used to estimate parameters. Out-of-sample validation via walk-forward or cross-validation is left for future work [Michaud and Michaud, 2008, Ch. 4].

10 Limitations

- (i) **Normality assumption.** Equations (5)–(6) use only the first two moments. Empirical equity returns exhibit excess kurtosis and negative skewness [Cont, 2001]. CVaR optimisation [Fabozzi et al., 2007, Ch. 8] provides a fat-tail robust alternative.
- (ii) **Estimation risk.** The sample covariance matrix $\hat{\Sigma}$ has $n(n+1)/2$ parameters estimated from T observations; for $n = 15$, $T = 986$, this is adequate, but for larger universes Ledoit-Wolf shrinkage [Ledoit and Wolf, 2004] is recommended.
- (iii) **Stationarity.** Sample moments estimated over 2021–2024 assume covariance stationarity. Regime-switching models [Hamilton, 1989] relax this assumption.
- (iv) **Survivorship bias.** Yahoo Finance data excludes delisted constituents, upward-biasing historical returns.
- (v) **Transaction costs.** Rebalancing frictions and market impact are not modelled.

11 Conclusion

This report has derived the core mathematical machinery of modern quantitative portfolio construction from first principles. Starting from the quadratic programme of Markowitz (MVO), we derived the efficient frontier analytically, established the Sharpe-tangency correspondence (19), and proved the full Bayesian Black-Litterman posterior (27) via completing-the-square. ESG and sector constraints were formulated

as (C-MVO), and attribution metrics (Jensen's α , β , IR, TE) were defined from the CAPM market model.

The resulting Python implementation achieves a Sharpe ratio of 1.840 on NIFTY 50 constituents, outperforming the benchmark by an estimated +16.53% alpha with an information ratio of 1.557—a strong in-sample result that motivates further out-of-sample validation.

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